La Scuola d'Italia / Huson / IB Math: Distributions First & last name:
26 February 2026 12th Grade

4.16 Do Now: Normal and Binomial Distribution Practice

1. The heights, H cm, of sunflowers grown in a garden competition are normally distributed with a mean of 148 cm and a standard deviation of 11 cm.

 (a) Find $\mathrm{P}(H < 140)$. [2]

 (b) Find $\mathrm{P}(135 < H < 160)$. [2]

 (c) The tallest 20% of sunflowers receive a prize. Find the minimum height, in cm, required to receive a prize. [2]

 (d) Given that a sunflower does not receive a prize, find the probability that it is taller than 150 cm. [2]

2. A city bus arrives on time at a particular stop with probability 0.65. Over a period of one week, the bus makes 20 scheduled arrivals. The number of times the bus arrives on time is denoted by X.

 (a) Find $\mathrm{P}(X = 13)$. [2]

 (b) Find $\mathrm{P}(X \geq 15)$. [2]

 (c) Find $\mathrm{E}(X)$ and $\mathrm{Var}(X)$. [2]

 (d) Find the greatest integer value of n such that $\mathrm{P}(X \geq n) > 0.95$. [3]

 (e) Explain in the context of the problem what the value of n found in part (d) represents. [1]